Adopt an airport project

London Heathrow Airport (IATA: LHR, ICAO: EGLL)

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Abstract: **Airports worldwide encounter daily capacity challenges which interrupt their scheduled operations and lead to delays. To address these issues efficiently, various Air Traffic Flow Management (ATFM) methods are applied to minimize delay and sustain efficient traffic flow.**

This report aims to examine the impact of several ATFM methods implemented at London Heathrow Airport (LHR) during a capacity reduction period. The objective is to determine the most effective approach to deal with the capacity limitations.

Keywords: capacity reduction, traffic regulation, Ground Delay Program (GDP), Ground Holding Problem (GHP), air delay, ground delay, intermodality

I. Introduction

Capacity challenges are a frequent feature of daily airport operations, mainly resulting from nominal capacity reductions produced by adverse weather conditions or technical issues, among other factors. Less frequently, they are due to an excess of demand. If no measure is implemented, these challenges produce delays that can escalate to inoperative values.

ATFM strategies deal with these situations applying different methods whose primary goal is to minimize the inefficiencies produced by the capacity issues.

This project focuses on London Heathrow Airport (IATA: LHR, ICAO: EGLL) as a case study for analyzing the impact of applying the mentioned methods in a European airport.

To begin with, the airport's data needed to perform the task is generated and a traffic regulation is implemented, causing a reduction in the capacity of the airport. After that, a Ground Delay Program (GDP) is designed and its repercussions are also examined, followed by the establishment of a Ground Holding Problem (GHP), which provides a different solution to the problematic scenario. Studying their impact provides metrics of paramount importance for the development of the project.

Ultimately, intermodality, which is an approach that considers alternative transportation modes, is evaluated to complement ATFM strategies.

A comprehensive comparison of the solutions previously obtained, weighing the advantages and limitations of each of them, provides a broader perspective of the situation. The most optimal solution to the problem, the one which minimizes delays and maintains an efficient flow of traffic, needs to be identified.

II. Hypothesis of the regulation

A. Data collection and computations

To begin the analysis, the flights arriving during the 29th and the 30th of January 2016 to London Heathrow Airport (LHR) appear within the data collected. Nevertheless, only the ones arriving during the former day are considered so as to analyze a period of exclusively 24 hours.

To continue, various key parameters which will be used through the whole development of the project are defined:

- Airport Acceptance Rate (AAR): maximum number of arrivals which an airport can safely handle. Also known as nominal capacity.

- Practical Airport Acceptance Rate (PAAR): realistic arrival rate which an airport can handle under current conditions. Also known, in this analysis, as reduced capacity.

- HFile: the hour when the regulation is officially filed and the regulated slots are provided to the airlines.

- HStart: the hour when the regulation takes effect and the airport begins operating at reduced capacity.

- HEnd: the hour when airport capacity returns to nominal values. Note that it is not the hour when the regulation ends.

- HNoReg: the hour when the regulation finishes.

The data needed to define the regulation is detailed below:

DATA TO DEFINE THE REGULATION:	
AAR [1]	45 flights/h
PAAR	25 flights/h
Slot duration in AAR	1,33 min (80 s)
Slot duration in PAAR	2,40 min (144 s)
HFile	6:00 h
HStart	8:00 h
HEnd	14:00 h

Table 1: Data to define the regulation.

The numerical results obtained when implementing the regulation are detailed in *VI. Analysis of results*.

B. *Graphical results*

The airport's incoming traffic flow during each hour of the day is presented in the following histogram.

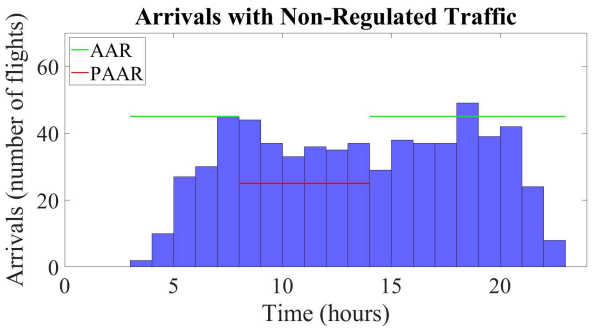

Figure 1: Histogram showing the arrivals with non-regulated traffic.

In the plot below, the aggregate demand as a function of time is represented as a continuous black line. The area between the curves equals the total delay of the regulation and, the intersection between the aggregate demand and the AAR lines provides HNoReg.

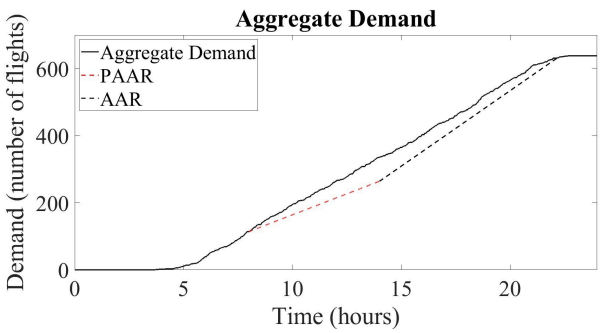

Figure 2: Plot showing the aggregate demand.

The following HNoReg value has been set in the project:

HNoReg	22:11 h

Table 2: HNoReg value.

Finally, the first slot allocation of the project is performed.

III. Ground Delay Program (GDP)

A. *Definition and computations*

In this section, a Ground Delay Program (GDP), defined as an ATFM method which delays aircraft at their departure airports to balance arrival demand with the capacity of the destination airport [2], is applied between HStart and HNoReg.

It only affects the non-exempt flights arriving during the period of time in which it is defined. The criteria for exempting a flight are based on the following aspects:

- Their origin airport is outside the radius of the GDP. The following radius value has been set in the project:

Radius	1000 NM (1852 km)

Table 3: Radius value.

- They are already flying when the GDP is defined (HFile).
- They come from an airport outside ECAC.

It must be mentioned that an extension period of 30 minutes has been considered in order not to penalize the flights which are ready to depart when the GDP is filed.

At this stage, a new slot allocation is performed. In contrast to the one performed when defining the regulation, priority has been given to the exempt flights.

According to the principles of this ATFM method, air delay is assigned to exempt flights and ground delay to the non-exempt ones.

An important parameter arises as a consequence of delay, the Controlled Time of Arrival (CTA), which is defined as the sum of the Scheduled Time of Arrival (STA) and the delay.

In addition, it has been considered a hypothetical situation in which the GDP is cancelled at HStart. The unrecoverable delay, defined as the delay incurred by the aircraft at the time of cancellation, is also computed.

B. *Graphical results*

The airport's incoming traffic flow when the GDP has been implemented is represented in the histogram attached below. As it can be observed, the excess flights during the reduced-capacity period have been allocated to the hours following the restoration of nominal capacity.

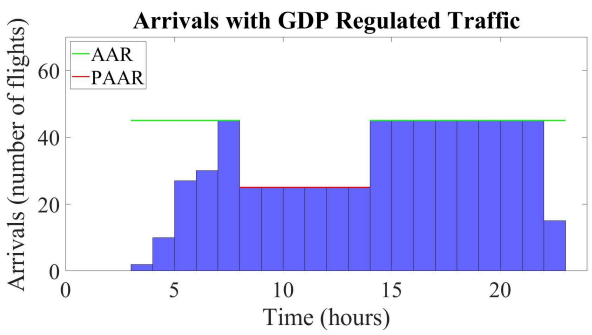

Figure 3: Histogram showing the arrivals with GDP regulated traffic.

C. *CO_2 emissions produced by delay*

In the following step, the CO_2 emissions due to delay have been calculated. On the one hand, for the emissions produced during ground delay, a value of 2,95 kg of fuel burnt per minute has been considered for all the aircraft categories [3], [4]. The total fuel consumption is obtained by multiplying the stated value by the ground delay performed by the non-exempt flights.

On the other hand, the air delay emissions have been computed defining an average value for the velocity of each aircraft category [6], [7], [8], [9] with which we can derive the distance traveled while performing air delay. Multiplying it by the corresponding average fuel consumption value on cruise [5], the fuel burnt is obtained.

DATA TO COMPUTE THE CO_2 EMISSIONS DUE TO AIR DELAY:			
Average velocity		*Average fuel consumption on cruise*	
Cat. A	900 km/h	Cat. A	13,78 kg/km
Cat. B	890 km/h	Cat. B	7,47 kg/km
Cat. C	860 km/h	Cat. C	5,70 kg/km
Cat. D	830 km/h	Cat.D	3,30 kg/km
Cat. E	770 km/h	Cat. E	3,24 kg/km
Cat. F	610 km/h	Cat. F	0,76 kg/km

Table 4: Data to compute the CO_2 emissions due to air delay.

Lastly, it has been considered that the combustion of 1 kg of aviation fuel produces 3,16 kg of CO_2 [10].

D. Effects of changing the GDP parameters

At this point, the GDP radius and HFile parameter will be changed in order to observe their effect on the ground and air delay, as well as on the CO_2 emissions.

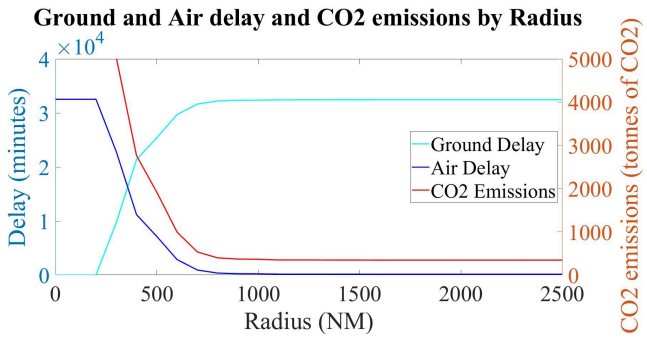

Figure 4: Ground and air delay and CO_2 emissions as a function of radius.

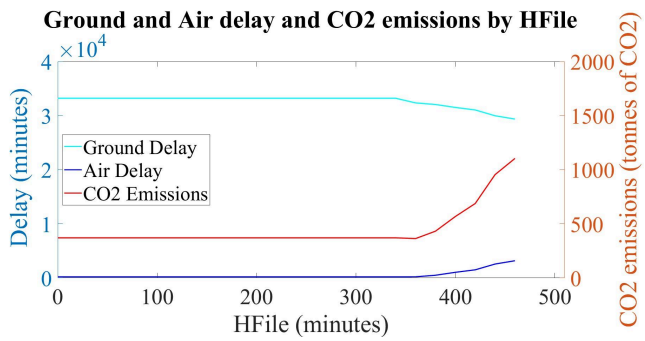

Figure 5: Ground and air delay and CO_2 emissions as a function of HFile.

As it can be observed in Figures 4 and 5, there is an inversely proportional relationship between the effect caused by the change in the radius parameter and HFile.

Firstly, as the radius increases, the ground delay grows and the air delay, as well as the CO_2 emissions produced by delay, decrease. This happens due to the fact that, as the GDP exemption radius becomes larger, more flights are no longer exempt and, therefore, are assigned ground delay. As one might imagine, fuel consumption on the ground, on average, is lower than in the air and, therefore, CO_2 emissions decrease.

Secondly, regarding the variation of HFile, the later the regulation is defined, the more flights are exempt from the GDP. Subsequently, the air delay increases. Following that, CO_2 emissions increase for the higher fuel consumption in the air, on average, compared to on the ground.

To continue, the evolution of the unrecoverable delay and the affected flights will be presented as a function of the same GDP parameters.

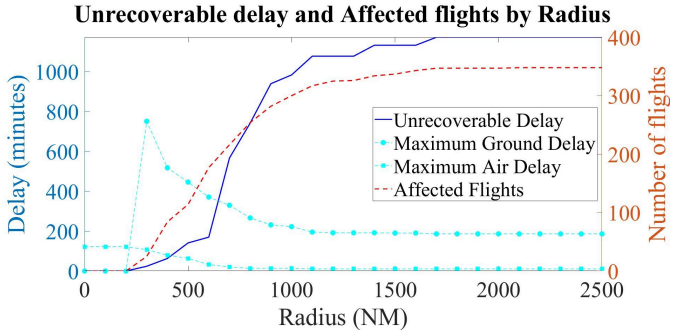

Figure 6: Unrecoverable delay, maximum ground and air delay and affected flights as a function of radius.

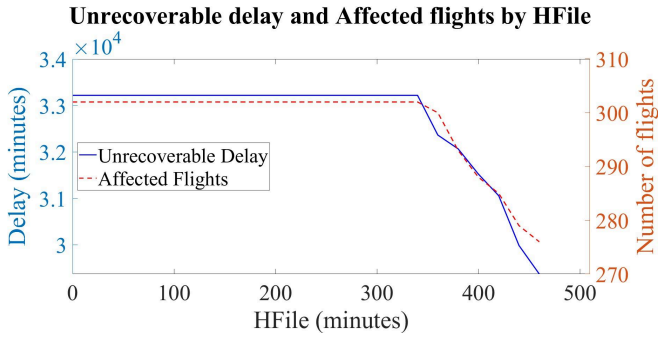

Figure 7: Unrecoverable delay and affected flights as a function of HFile.

Figure 6 relates the increase of radius with the growth of the number of affected flights, as it has been previously mentioned. Therefore, unrecoverable delay also grows due to the fact that more flights perform ground delay before the regulation is hypothetically cancelled.

Moreover, in this figure, two extra KPIs have been added: maximum ground and air delay. The peak in the maximum ground delay representation corresponds to the lowest value of radius for which the GDP affects a flight. To be concise, there is a single non-exempt flight which is placed after the exempt flights, which have priority, and assumes the highest value of ground delay. As radius increases, once again, more flights are affected by the GDP and are assigned ground delay. When it comes to air delay, the opposite scenario is shown.

In Figure 7, as HFile increases, more flights are exempt and, if the regulation was cancelled, less unrecoverable delay

would have occurred. It is worth mentioning that the unrecoverable delay has a constant value until the first regulated flight departs from its origin airport. Subsequently, defining HFile earlier would have no effect because the number of flights affected by the regulation does not vary.

Having said that, the optimal values for the radius and the HFile parameters (which are already used since the beginning of the project) are 1000 NM and 6:00 h, respectively.

Both parameters are within the interval in which ground delay is maximized and air delay is minimized (Figures 4 and 5). Furthermore, for them, the unrecoverable delay does not have a very high value and there is a reasonable number of flights affected by the GDP (Figures 6 and 7).

E. Cancellation and substitution

Next, it has been considered that the airline British Airways cancels all the flights with more than 3 hours of delay. A second slot allocation within the GDP has been performed.

To continue, substitution has been implemented within the flights of the same airline and finally, to occupy the remaining empty slots, compression is performed.

The new slot allocations and delays are presented in the following plots.

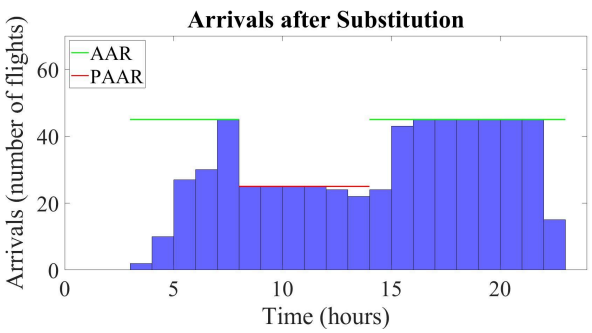

Figure 8: Histogram showing the arrivals when applying substitution.

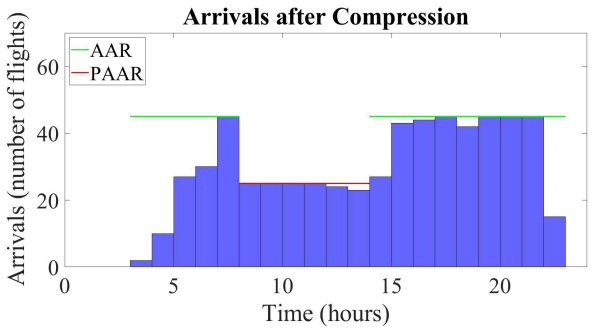

Figure 9: Histogram showing the arrivals when applying compression.

Comparing Figures 8 and 9, the effect produced by substitution and the following compression can be perceived. During the first hour in which nominal capacity is recovered, considering only substitution, 24 flights are scheduled to arrive at the airport. Nevertheless, when compression is applied, this value ascends to 27. The remaining hours of the journey also witness this aspect through a slight variation in arrivals.

F. Cost of delay

To conclude this section, the cost of delay obtained implementing the GDP has been calculated. Note that the cost parameter which has been used is defined in *IV. Ground Holding Problem (GHP) – B. Cost of delay*.

The numerical results obtained when implementing the GDP are detailed in *VI. Analysis of results*.

IV. Ground Holding Problem (GHP)

A. Definition and computations

Throughout this section, a Ground Holding Problem (GHP) is applied to optimize aircraft arrival scheduling at the airport and, subsequently, find the best arrival times.

The regulation and affected flights from the GDP, as well as the slot allocation, are used to establish this ATFM method.

Before its implementation, some aspects must be defined:

- A flight cannot arrive before its ETA and, therefore, a large cost has been assigned to the slots present before its arrival.
- Each slot can accommodate only one flight *($b_t = 1$)*:

$$\sum_f x_{f\,t} \leq b_t \tag{1}$$

- Each flight can be assigned to a slot only once:

$$\sum_{t \geq e(f)} x_{f\,t} = 1 \tag{2}$$

The cost function which has to be minimized is presented below:

$$min \sum_f \sum_t c_{f\,t} x_{f\,t} \tag{3}$$

Where:

$$c_{f\,t} = r_f (t - e(f))^{(1+\varepsilon)} \tag{4}$$

$t = set\ of\ slots$

$f = set\ of\ flights$

$b_t = constrained\ airport's\ arrival\ capacity\ at\ slot\ t$

$r_f = cost\ parameter\ assigned\ to\ flight\ f$

$e(f) = STA\ of\ flight\ f$

$c_{f\,t} = cost\ of\ assigning\ flight\ f\ to\ arrive\ at\ slot\ t$

$x_{f\,t} = 1\ if\ flight\ f\ is\ assigned\ to\ slot\ t,\ 0\ otherwise$

B. Cost of delay

The cost parameter (in 2014-Euros) is derived from the full tactical cost of delay, which includes the costs for fuel and maintenance, the expenses for both the fleet and the crew, the passenger costs and the reactionary effects. Depending on whether the flight is assigned ground or air delay, at-gate or en-route base scenario values are used, respectively.

Moreover, for categories A to F, aircraft models A388, B744, B763, A320, E190 and ZZZZ have been utilized, respectively. For categories A and F, as they do not appear tabulated, the cost has been derived from the cost per seat obtained in categories B and E, correspondingly.

The selection of the cost parameter r_f is based on the data provided in [11], which is detailed in *Annexes – A. Cost of delay*.

It is important to note that the longer the delay, the higher the assigned cost. While short delays are acceptable, large delays are infeasible. In particular, the costs associated with flights performing on-time performance (OTP), that indicates the total number of flights which arrive on time or, at most, within 15 minutes of its scheduled arrival, are relatively low.

Consider that, for air delay values exceeding 210 minutes (3,5 h), a large cost has been assigned in order to penalize them when performing the GHP slot allocation.

Attention should be drawn to the fact that the base scenario values include an average load factor (LF) of 75 % for narrow-body aircraft and 80 % for wide-body ones.

Finally, a cost associated with connecting passengers has been added. To begin with, in 2019, 34 % of the passengers in LHR did a transfer [12]. This year has been selected because aviation had not yet been affected by the impacts of the COVID-19 pandemic and it can be considered a base scenario.

To calculate this cost, 34 % of the passengers of each flight are selected and the values are based on the scenario where the transfer cannot be performed. If the flight accumulates at least 180 minutes of delay, the cost per transfer passenger is determined according to the distance flown (in Euros) [13].

COST PER TRANSFER PASSENGERS:			
Distance	0 – 1500 km	> 1500 km	> 3500 km
Cost per pax	250 €	400 €	600 €

Table 5: Cost per transfer passenger according to the distance flown by the flight.

After that, the GHP is implemented and, based on the costs per flight, an optimal slot allocation is obtained. Furthermore, the total delay cost is calculated. See below the histogram showing the arrivals at LHR when a GHP is implemented.

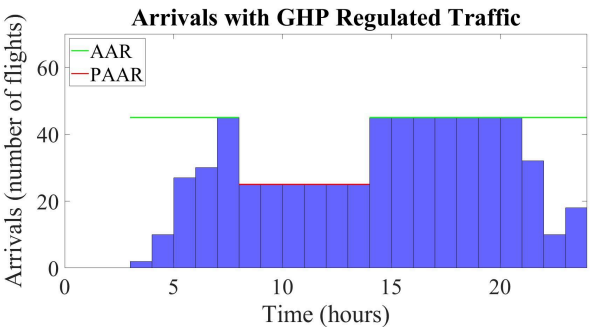

Figure 10: Histogram showing the arrivals with GHP regulated traffic.

C. CO_2 emissions produced by delay

To continue, the CO_2 emissions produced by delay are calculated. The reference values used for the computations are the ones presented in *III. Ground Delay Program (GDP) – C. CO_2 emissions produced by delay*.

The numerical results obtained when implementing the GHP are detailed in *VI. Analysis of results*.

V. Intermodality

A. Definition and computations

This section focuses on the implementation of intermodality, which involves substituting air travel with, in this case, train journeys. It is performed mainly, when travel times are acceptable, to minimize the arrival demand at the airport and reduce the CO_2 emissions produced by aviation. Its application is aimed to determine if the regulation could have been avoided or if the total delay decreases.

From this point forward, the concept of Door-to-Door travel time (D2D), defined as the total time required to travel from the starting point to a final destination, will be of great importance. It includes not only the time spent during the journey, whether by plane or train, but also the time spent reaching the station, departing from it and completing security controls, known as additional travel time (ATT).

Initially, 6 airports located within a 200-600 km radius of LHR and with a direct train connection with London of no more than 4-6 hours have been selected [14], [15]. Moreover, it has been considered that all the flights, being short range, are operated with Cat. D aircraft, specifically the A320. Using this aspect, the number of seats is obtained (165 seats) and this value is utilized to derive the CO_2 emissions per passenger.

The data needed to analyse each particular case within the regulation period of time, which is presented in the table attached below, comes from: flight time [16], high-speed rail time [14], [15], flight and high-speed rail ATT [17], CO_2 emissions [18] and distances [19].

DATA TO ANALYSE INTERMODALITY:				
Origin	*Flight time*	*Flight ATT[1]*	*HSR[2] time*	*HSR ATT*
Paris	1 h 25 min	2 h 15 min	2 h 17 min	1 h
Brussels	1 h 10 min	2 h 16 min	1 h 56 min	52 min
Amsterdam	1 h 25 min	2 h 43 min	3 h 57 min	57 min
Manchester	1 h 25 min	2 h 38 min	2 h 35 min	1 h 7 min
Edinburgh	1 h 40 min	2 h 27 min	4 h 11 min	49 min
Newcastle	1 h 25 min	2 h 42 min	2 h 36 min	56 min
1. ATT → Additional Travel Time 2. HSR → High-Speed Rail				

Table 6: Data to analyse intermodality.

B. *Comparison of options*

To continue, D2D travel times and total CO_2 emissions for both air and rail options are compared in two different scenarios.

Remember that the D2D time is calculated with the sum of either the flight or the HSR time and the corresponding additional time (Table 6).

- Scenario 1: flight times without delay are considered.

SCENARIO 1:				
Origin	*Flight D2D time*	*HSR D2D time*	*Flight CO_2 emissions [kg/pax]*	*HSR CO_2 emissions [kg/pax]*
Paris	3 h 40 min	3 h 17 min	126,3	1,7
Brussels	3 h 26 min	2 h 48 min	121.1	2
Amsterdam	4 h 8 min	4 h 54 min	123,5	11,1
Manchester	4 h 3 min	3 h 42 min	128,7	15,9
Edinburgh	4 h 7 min	5 h	118,3	32,8
Newcastle	4 h 7 min	3 h 32 min	103	23,2

Table 7: Comparison of options - Scenario 1.

To start, the D2D travel times are compared. According to the data provided in Table 7, in 4 of the 6 cases presented the train option proves faster than flying (29,25 minutes faster, on average), especially from Brussels and Newcastle, where the HSR trips are 38 and 35 minutes shorter, respectively. Nevertheless, for the two remaining cases, flying is the fastest option, with an average time advantage of 49,5 minutes.

When it comes to the CO_2 emissions per passenger, the HSR option is considerably more environmentally beneficial than flying, with an average reduction of 105,7 kg/pax.

Consequently, the HSR could be selected as the optimum option. It offers similar or faster travel times on most routes and is more sustainable than flights. This tendency is shown in the plot below.

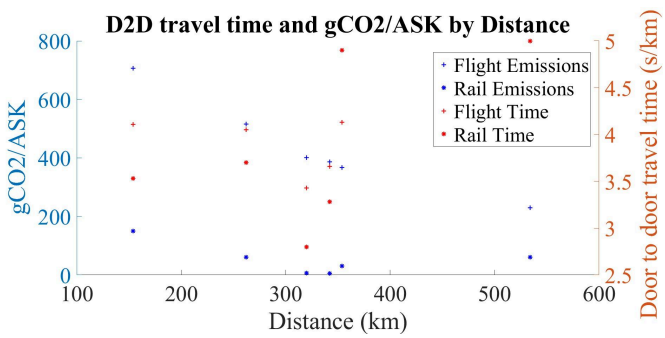

Figure 11: Plot showing the D2D travel time and gCO_2/ASK as a function of distance.

- Scenario 2: flight times with delays derived from the GDP are considered.

It must be taken into consideration that these flights are always assigned ground delay due to the fact that they meet all the conditions not to be exempt from the GDP.

Furthermore, as several flights departing from these airports arrive at LHR within the regulation period, an average ground delay has been considered for each of the six cases.

SCENARIO 2:				
Origin	*Flight D2D time → Average ground delay used*	*HSR D2D time*	*Flight CO_2 emissions → Ground delay CO_2 em. [kg/pax]*	*HSR CO_2 emissions [kg/pax]*
Paris	5 h 28 min → 1 h 48 min	3 h 17 min	132,4 → 6,1	1,7
Brussels	5 h 33 min → 2 h 7 min	2 h 48 min	128,3 → 7,2	2
Amsterdam	6 h 3 min → 1 h 55 min	4 h 54 min	130,1 → 6,6	11,1
Manchester	5 h 58 min → 1 h 55 min	3 h 42 min	135,3 → 6,6	15,9
Edinburgh	5 h 25 min → 1 h 18 min	5 h	122,8 → 4,5	32,8
Newcastle	5 h 47 min → 1h 40 min	3 h 32 min	108,7 → 5,7	23,2

Table 8: Comparison of options - Scenario 2.

As it can be observed, in this second scenario in which ground delay is considered, the HSR option is always faster than the flight. Train trips are, on average, 110,2 minutes shorter than flights. The greatest time reduction is achieved when travelling from Brussels, where 2 h and 45 min can be saved by taking the train.

When considering the CO_2 emissions, it ought to be mentioned that, due to the ground delay applied, emissions increase. Nevertheless, the impact of delay is relatively small in comparison with the en-route flight emissions, which are, as it has been previously mentioned in Scenario 1, significantly higher than the HSR ones. There is an average reduction of 111,8 kg/pax in emissions when travelling by train.

To conclude, HSR remains the most feasible solution when travelling short distances. Its significant reduction in travel time, especially when delay is considered, combined with the small amount of emissions it produces, makes the train the most effective alternative to flights.

C. Implementation of GDP and GHP

At this stage, the flights departing from the airports previously considered within the regulation time interval (62 flights) have been removed from the original arrival demand at LHR to implement a GDP and a GHP using this reduced set of flights.

The most noticeable impact of implementing intermodality is the obtention of the new HNoReg parameter.

Figure 12 shows the aggregate demand computed with the reduced set of flights. The HNoReg parameter, obtained in the intersection between the aggregate demand and the AAR lines, has a lower value, indicating an earlier ending time of the regulation.

The new HNoReg parameter has the following value:

HNoReg	17:23 h

Table 9: HNoReg value when implementing intermodality.

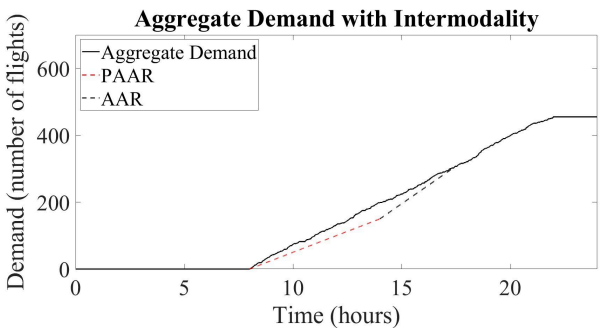

Figure 12: Plot showing the aggregate demand when implementing intermodality.

The numerical results obtained when implementing intermodality are detailed in *VI. Analysis of results.*

VI. Analysis of results

KPIs	INITIAL REGULATION	
	Base scenario	**Intermodality**
CAPACITY (KPA)		
Number of flights affected	518 flights	302 flights
Total delay of the regulation	32259,24 min	13802,35 min
Average delay per flight	62,28 min	45,70 min
SD[1] of delay	68,23 min	59,72 min
RSD[2] of delay	108,45 %	129,02 %
Maximum delay	223 min	168 min
1. SD → Standard deviation 2. RSD → Relative standard deviation		

Table 10: Regulation results.

KPIs	GDP		GHP	
	Base scenario	**Intermodality**	**Base scenario**	**Intermodality**
COST EFFECTIVENESS AND OPERATIONAL EFFICIENCY (KPAs)				
Flights affected by air delay	218 flights	160 flights	218 flights	160 flights
Total air delay	228 min	209,33 min	4776,93 min	5171,33 min
Average air delay per flight	1,05 min	1,31 min	21,91 min	32,32 min
SD of air delay	2,13 min	2,41 min	28,93 min	27,14 min
RSD of air delay	203,31 %	184,11 %	132,02 %	83,97 %
Maximum air delay	13,13 min	13,13 min	207,33 min	87,33 min
Flights affected by ground delay	300 flights	142 flights	300 flights	142 flights

Total ground delay	32361,13 min	13766,80 min	29904,67 min	9552,60 min
Average ground delay per flight	107,87 min	96,95 min	99,68 min	67,27 min
SD of ground delay	56,76 min	52,16 min	214,34 min	106,84 min
RSD of ground delay	52,62 %	53,80 %	215,02 %	158,82 %
Maximum ground delay	222,53 min	168,33 min	952,67 min	952,67 min
Air/Ground delay ratio	0,0070	0,0152	0,1597	0,5414
On-time performance (OTP)	225 flights	163 flights	218 flights	83 flights
Unrecoverable delay	983,53 min	913,73 min	866,53 min	791,67 min
Implementation of cancellation, substitution and compression:				
Cancelled flights	27 flights	0 flights	6 flights	1 flight
Total delay	26369 min	13979 min	16901 min	12879 min
Average delay per flight	53,70 min	46,29 min	33,94 min	42,93 min
Maximum delay	214 min	168 min	207 min	117 min
BAW air delayed flights	40 flights	34 flights	67 flights	58 flights
Total air delay of BAW flights	100,87 min	93,87 min	2183,40 min	2219,47 min
Average air delay per BAW flight	2,52 min	2,76 min	32,59 min	38,27 min
BAW ground delayed flights	115 flights	78 flights	138 flights	75 flights
Total ground delay of BAW flights	10541,47 min	7870,27 min	6195,07 min	4362,07 min
Average ground delay per BAW flight	91,66 min	100,90 min	44,89 min	58,16 min
Average decrease of total delay for the remaining BAW flights	37,87 min	0 min	24,21 min	6,74 min
Average decrease of air delay for the remaining BAW flights	0 min	0 min	0 min	0 min
Average decrease of ground delay for the remaining BAW flights	19,98 min	0 min	35,30 min	11,17 min
Computation of costs:				
Total cost of delay	5987645 €	3578194 €	2986202€	1186238€
Average cost of total delay per flight	12194,80 €	11848,32 €	5996,39 €	3954,13 €
ENVIRONMENT (KPA)				
Total CO_2 emissions	364446,78 kg	186638,23 kg	1450689,36 kg	1428346,16 kg
Avg. CO_2 emissions per flight	703,57 kg	618,01 kg	2800,56 kg	4729,62 kg
Air delay CO_2 emissions	62776,30 kg	58304,12 kg	1171918,06 kg	1339296,82 kg
Avg. air delay CO_2 emissions per flight	287,96 kg	364,40 kg	5208,52 kg	8267,26 kg
Ground delay CO_2 emissions	301670,48 kg	128334,11 kg	278771,30 kg	89049,34 kg
Avg. ground delay CO_2 emissions per flight	1005,57 kg	903,76 kg	951,44 kg	636,07 kg

Table 11: GDP and GHP results.

Firstly, a comparison between the outcomes of the regulation in a base scenario and when implementing intermodality, presented in Table 10, is performed. To begin with, 518 flights are within the regulation time interval and, when intermodality is applied, this number is reduced significantly to 302 flights. This reduced set of flights is obtained from the substitution of 62 flights with HSR trips and the acquisition of a lower HNoReg parameter.

Subsequently, intermodality also decreases the total delay of the regulation by 57,2 % compared to the base scenario, showing efficiency minimizing delays. The average delay per flight and the maximum delay are also lower. Note that the latter KPI is reduced by 24,7 %, which represents an improvement in the worst-case delay. The reduction in the SD reveals a decrease in the dispersion of delays and the increase in the RSD suggests that the relative variability compared to the average delay is higher.

Secondly, the Ground Delay Program (GDP) and the Ground Holding Problem (GHP) results, detailed in Table 11, are compared within the base scenario. In the first place, even though the number of flights affected by air and ground delay is equal in both ATFM methods, the GHP experiences higher total, average and maximum air delay, as well as more unpredictable values, compared to the GDP. Regarding ground delay, the GDP shows greater total and average values, but the maximum appears in the GHP, which also has the highest dispersion of ground delay. Finally, the GDP outperforms the GHP in OTP, with more flights arriving on time, but presents a higher unrecoverable delay.

Having said that, it can be inferred that the GDP tends to have more consistent and higher ground delays, added to a better on-time performance, which is coherent with our objective. On the contrary, the GHP has the most extreme values (the largest maximum delays) and the highest total air delay, which needs to be minimized.

It can be verified that, even though the total delay obtained in the GDP and the GHP is slightly different, the values are so similar that it can be considered independent of the method applied and depends exclusively on the parameters defining the regulation. The small difference can be attributed to the reordering of flights within the slots, as each slot is several minutes long.

At this point, the implementation of intermodality is taken into consideration. This scenario results in fewer flights being affected by delay. To compare the results, the average values for delay per flight are considered. It can be noticed that for both methods, intermodality increases the average air delay and decreases the average ground delay per flight, which goes against our fundamental objective. OTP is considerably worsened and unrecoverable delay slightly improved.

With the results obtained, it cannot be stated that with intermodality it is possible to prevent the regulation defined. Nevertheless, it substantially reduces the total delay.

To continue, cancellation, substitution and compression is applied to the scenarios previously mentioned. Depending on the ATFM method applied (complemented or not with intermodality), the number of flights cancelled is different. This aspect arises from the fact that, depending on the slot allocation performed, a specific number of flights meet the requirements to be cancelled.

It is important to highlight that the GDP is the method in which more flights are cancelled. Moreover, it is the one which has more ground than air delayed BAW flights (there is a difference of 75 flights), even though this tendency appears in all the cases.

In contrast to the GHP, the GDP experiences a clear positive contribution when applying intermodality, with which few or no flights are cancelled. Ground delay continues being higher than air delay even though there is a slight increase in the average air delay per flight.

Lastly, with the base scenario GDP, we obtain the highest average decrease of total delay for the remaining BAW flights.

Regarding the economic aspect, the GDP is 100,51 % more expensive than the GHP. With intermodality, costs are reduced, especially in the GDP, and the GHP continues being the cheapest method.

Finally, in relation to the CO_2 emissions, the GDP is the most environmentally friendly method. Most of its emissions are released during ground delay and, therefore, pollution is less than performing air delay. As for intermodality, even though it produces a slight increase in the average air delay CO_2 emissions when complementing the GDP, it achieves a major reduction of the total emissions, which is a critical aspect in this analysis due to the significant relationship between the environment and aviation.

A. Identification of Key Performance Areas (KPAs)

The KPIs presented in Tables 10 and 11 have been classified in the following KPAs [20]:

- Capacity: it contains the KPIs which define the regulation prior to the implementation of any ATFM method. Since no solution is applied at this point, the results show how the airport's capacity responds to a capacity reduction period.

- Cost Effectiveness and Operational Efficiency: they include all the KPIs which arise from the implementation of the GDP and the GHP, excluding CO_2 emissions.

The explanation lies in the fact that, as it has been stated at the beginning of this report, the main objective of implementing ATFM methods is to minimize the inefficiencies produced by the capacity reduction period. In other words, the aim is to be as efficient as possible in flight operations and ensure an optimal cost when establishing the GDP and the GHP.

- Environment: it has the KPIs related to the CO_2 emissions produced by delay. It is evident that there is a close relationship between the emissions produced by aviation and the preservation of the ecosystem.

VII. Conclusions

Once the different scenarios which aim to manage the capacity issues derived from the regulation have been assessed, an optimal solution must be selected.

It has been considered that the implementation of a Ground Delay Program (GDP) complemented with intermodality is the best approach. The reason lies, primarily, in the fact that it maximizes ground delay and minimizes air delay, which is the aim of the ATFM strategies. This aspect can be demonstrated with the ratio between air and ground delay, which has the smallest value. Regarding intermodality, it reduces the total delay efficiently.

On the contrary, GHP does not properly optimize the distribution of delays when comparing it to the GDP (the ratio is nearly 23 times bigger) and intermodality, in this case, does not enhance the situation.

Furthermore, even though the GHP is cheaper than the GDP, it has demonstrated that it is not efficient enough to justify its use only because of an economic reason. As a result, the cost of the GDP, despite being more expensive, is assumed. Intermodality plays a key role reducing the expense of its implementation.

To conclude, when it comes to the CO_2 emissions and the premise of protecting the environment arises, the GDP with intermodality greatly surpasses all the remaining options in releasing the fewest emissions.

Contributions

The contributions of each member of the group to the different work packages which constitute the project are detailed below. The most important functions created within the coding environment are specified:

WP1. Joel: main programmer (aggregateDemandIn Seconds, computeSlots). Pau: programming support (assign Category, verify_ECAC_member) and data research. Àlex: programming support (plot_flights_histogram, assignPAX).

WP2. Joel: main programmer (computeAircraftStatus, assignSlotsGDP, assignGround_Air_Delays, CTA_Histogram, computeCO2Emissions), plots development and cancellations. Pau: programming support (computeAircraftStatus, Compute UnrecoverableDelay) and data research. Àlex: programming support (computeAircraftStatus, computeCTA, ComputeDelay InRegulation) and short report development.

WP3. Joel: programming support (computeTask2, obtain_r _Parameter, obtainGDPCosts). Pau: programming support (obtain_r_Parameter) and data research. Àlex: main programmer (computeTask2, obtain_r_Parameter) and LP problem definition.

WP4. Joel: main programmer (computeCO2EmissionsOfA Flight, plotFuelConsumptionByCategoryAndKm). Pau: main researcher (intermodality and sustainability, D2D times, CO_2 computations) and programming support (ComputeDelayFor OriginAirport, CancelIntermodalityFlights). Àlex: analysis of results obtained when implementing intermodality.

WP5. Joel: analysis of results. Pau: analysis of results. Àlex: analysis of results and plots, identification of KPIs and KPAs and final report development.

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A. Cost of delay

Table 26. AT-GATE / BASE / full tactical costs

Delay (mins)	5	15	30	60	90	120	180	240	300
B733	70	430	1 550	7 020	19 160	36 220	49 040	66 480	89 310
B734	80	480	1 740	7 930	21 690	40 960	55 340	74 780	100 040
B735	70	390	1 400	6 280	17 110	32 350	43 900	59 720	80 590
B738	90	540	1 940	8 860	24 270	45 750	61 740	83 220	110 920
B752	100	620	2 290	10 620	29 250	55 150	74 240	99 700	132 200
B763	170	900	3 200	14 780	39 960	85 300	121 880	152 860	191 990
B744	240	1 370	5 000	23 430	63 710	136 330	194 330	242 440	302 200
A319	70	440	1 600	7 320	20 040	37 850	51 240	69 420	93 180
A320	80	500	1 820	8 350	22 920	43 250	58 420	78 890	105 380
A321	100	580	2 160	10 010	27 580	51 990	70 060	94 250	125 240
AT43	30	180	610	2 610	6 960	13 290	18 550	26 360	37 610
AT72	40	240	820	3 600	9 690	18 430	25 380	35 350	49 210
DH8D	40	250	890	3 900	10 530	19 990	27 480	38 120	52 780
E190	60	320	1 150	5 140	13 970	26 440	36 060	49 420	67 340
A332	180	990	3 550	16 480	44 620	95 330	136 120	170 480	213 660

With reactionary costs.

Table 12: AT-GATE full tactical costs [11].

Table 28. EN-ROUTE / BASE / full tactical costs

Delay (mins)	5	15	30	60	90	120	180	240	300
B733	250	960	2 620	9 160	22 370	40 490	55 440	75 020	99 980
B734	260	1 010	2 800	10 060	24 890	45 220	61 730	83 300	110 690
B735	230	880	2 370	8 230	20 030	36 250	49 750	67 510	90 340
B738	270	1 080	3 030	11 040	27 540	50 120	68 290	91 960	121 840
B752	340	1 340	3 730	13 500	33 570	60 910	82 870	111 200	146 580
B763	510	1 920	5 240	18 860	46 080	93 460	134 120	169 180	212 390
B744	930	3 440	9 140	31 710	76 130	152 890	219 170	275 560	343 600
A319	240	950	2 620	9 350	23 090	41 910	57 330	77 540	103 320
A320	250	1 010	2 850	10 410	26 000	47 360	64 590	87 110	115 660
A321	300	1 190	3 380	12 450	31 240	56 880	77 390	104 020	137 460
AT43	70	290	830	3 040	7 600	14 140	19 840	28 080	39 750
AT72	90	390	1 130	4 210	10 620	19 650	27 220	37 810	52 280
DH8D	110	450	1 280	4 690	11 720	21 570	29 840	41 270	56 710
E190	180	700	1 910	6 670	16 250	29 490	40 630	55 510	74 950
A332	570	2 140	5 860	21 100	51 560	104 580	149 980	188 960	236 770

With reactionary costs.

Table 13: EN-ROUTE full tactical costs [11].